FORWARD DEPLOYED ENGINEER PORTFOLIO

Inventory Management System

Technical Portfolio & Architecture Specification

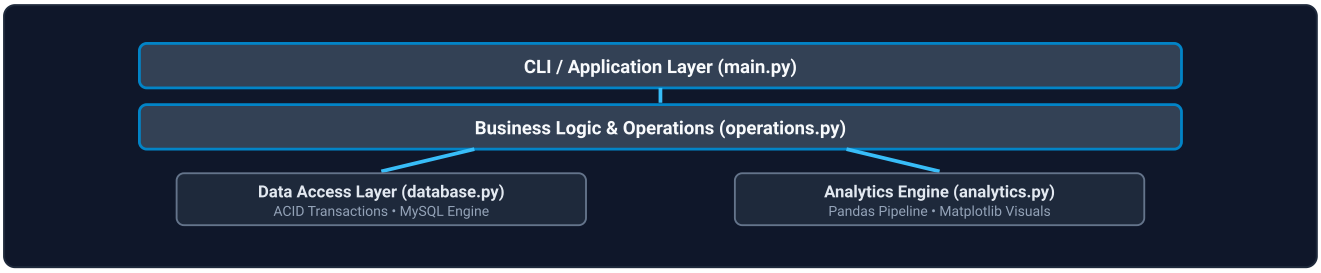
ROLE TARGET	DOMAIN	TECH STACK
Forward Deployed Engineer (FDE)	**Enterprise Logistics & Supply Chain**	**Python, MySQL, Pandas, Matplotlib**

1. System Overview & Problem Statement

Modern retail and logistics networks require deterministic, transactional accuracy for high-volume stock movements. Conventional spreadsheet-based tracking fails to provide transaction isolation, leading to race conditions, inventory drift, and inaccurate demand forecasting.

This **Inventory Management System** addresses these operational bottlenecks by providing transactional guarantees, automated threshold auditing, and integrated analytical pipelines.

2. Technical System Architecture



3. Key Architecture Principles

- **ACID-Compliant Transactions:** Enforces strict isolation and atomicity across stock additions, order fulfillments, and supplier returns.
- **Analytical Intelligence:** Integrated data pipelines using Pandas and Matplotlib for demand analytics and velocity mapping.
- **Auditability & Traceability:** Immutable logging of stock updates and dynamic reorder threshold alerts.

4. Database Schema Design (DDL)

```
CREATE DATABASE IF NOT EXISTS inventory_db; USE inventory_db; CREATE TABLE IF NOT EXISTS suppliers ( supplier_id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(100) NOT NULL, contact_email VARCHAR(100) UNIQUE, phone VARCHAR(20) ); CREATE TABLE IF NOT EXISTS products ( product_id INT AUTO_INCREMENT PRIMARY KEY, sku VARCHAR(50) UNIQUE NOT NULL, name VARCHAR(100) NOT NULL, category VARCHAR(50), unit_price DECIMAL(10, 2) NOT NULL, stock_quantity INT DEFAULT 0, reorder_threshold INT DEFAULT 10, supplier_id INT, FOREIGN KEY (supplier_id) REFERENCES suppliers(supplier_id) ON DELETE SET NULL ); CREATE TABLE IF NOT EXISTS inventory_transactions ( transaction_id INT AUTO_INCREMENT PRIMARY KEY, product_id INT NOT NULL, transaction_type ENUM('INBOUND', 'OUTBOUND', 'ADJUSTMENT') NOT NULL, quantity INT NOT NULL, timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP, FOREIGN KEY (product_id) REFERENCES products(product_id) );
```

5. Engineering Practices & Execution Strategy

- **Separation of Concerns:** Modular architecture isolates database connectivity, business rules, and analytical reporting into dedicated sub-modules.
- **Secure Environment Configuration:** Decoupled sensitive database credentials using standardized environment variable parameters (`.env.example`).
- **Data Integrity & Constraints:** Foreign key constraints and atomic transaction handling prevent partial writes during concurrent order processing.
- **Automated Reorder Alerts:** Threshold detection logic dynamically triggers system alerts when SKU quantities drop below safe operational bounds.